

Notes: Almazan, Brown, Carlson, and Chapman (JFE, 2004)

Why Constrain Your Mutual Fund Manager?

Contents

1	One-sentence summary	3
2	Big picture	3
2.1	Research question	3
2.2	Core answer	3
2.3	Main empirical findings	3
2.4	Economic takeaway	3
3	Incremental contribution of the paper	3
3.1	Direct evidence on mutual fund investment restrictions	3
3.2	Contracting interpretation of constraints	4
3.3	Joint analysis of multiple monitoring mechanisms	4
3.4	Evidence against a simple performance-cost view	4
3.5	Methodological contribution	4
4	Data and measurement	4
4.1	Sample and sources	4
4.2	Six investment restrictions	4
4.3	Constraint score	5
4.4	Alternative monitoring variables	5
4.5	Performance measures	5
5	Empirical specifications	5
5.1	Constraint-score regressions	5
5.2	Individual-restriction probits	6
5.3	Performance tests	6
6	Identification and empirical design	6
6.1	Nature of identification	6
6.2	Optimal-contracting prediction	6
6.3	Performance implication	6
6.4	Main threats	6
7	Tables and figures	7
7.1	Table 1: Incidence of constrained and unconstrained funds; Table 2: Use of investment techniques by unconstrained funds	7
7.2	Table 3: Pooled regression estimates	8
7.3	Table 4: Summary tests from individual constraint pooled probits	9

7.4	Table 5: Returns to a long-short strategy based on score; Table 6: Returns to a long-short strategy based on absolute errors	10
7.5	Table 7: Regression of fund returns on constraint variables	11
8	Short summary	12
9	Conclusion	13
9.1	Core conclusion	13
9.2	Economic intuition	13
9.3	Incremental contribution revisited	13
9.4	Performance implications	13
9.5	Limitations	13
9.6	Takeaway	13

1 One-sentence summary

Mutual fund investment restrictions are widespread, systematically related to alternative monitoring mechanisms, and do not predict worse risk-adjusted performance, supporting an optimal-contracting view of constraints in delegated portfolio management.

2 Big picture

2.1 Research question

Why do investors hire professional mutual fund managers but then constrain their investment choices through prohibitions on borrowing, margin, short sales, derivatives, futures, and restricted securities?

2.2 Core answer

Investment restrictions can be understood as one contractual mechanism for controlling agency problems between fund shareholders and portfolio managers. Constraints are used more when alternative monitoring mechanisms are weaker or less effective.

2.3 Main empirical findings

- Constraints are common but not uniform across funds or time.
- Margin restrictions are almost universal, while restrictions on options, futures, restricted securities, borrowing, and short sales vary substantially.
- Funds with more inside directors, team management, more experienced managers, and smaller fund-family monitoring are more constrained.
- Funds belonging to larger complexes and funds with more independent boards are less constrained.
- Low-constraint and high-constraint funds earn similar abnormal returns, and strategies based on constraint intensity do not generate significant alpha.

2.4 Economic takeaway

The evidence supports an optimal contracting interpretation: constraints are not simply inefficient rules that handicap talented managers. They are chosen as part of an equilibrium governance package that trades off managerial flexibility against monitoring and agency costs.

3 Incremental contribution of the paper

3.1 Direct evidence on mutual fund investment restrictions

Previous mutual fund research focused mostly on fees, incentives, flows, and performance. This paper studies the investment restrictions embedded in fund policy statements and Form N-SAR disclosures, making constraints themselves the object of analysis.

3.2 Contracting interpretation of constraints

The paper treats restrictions as contractual governance tools, analogous to debt covenants. This shifts the question from “do constraints reduce returns?” to “when are constraints optimally used as substitutes for other monitoring mechanisms?”

3.3 Joint analysis of multiple monitoring mechanisms

The paper links constraints to four monitoring channels:

- board monitoring;
- labor-market monitoring and manager career concerns;
- peer monitoring through fund complexes;
- product-market monitoring through load-fee structure.

This broader governance perspective is the paper’s main conceptual contribution.

3.4 Evidence against a simple performance-cost view

If constraints were merely costly limits on manager skill, constrained funds should underperform. The paper finds no significant performance difference between high- and low-constraint funds, making the optimal-contracting interpretation more plausible.

3.5 Methodological contribution

The paper builds a constraint score summarizing six specific restrictions and estimates both pooled regressions and individual probits. It therefore combines descriptive institutional measurement with cross-sectional and time-series econometric evidence.

4 Data and measurement

4.1 Sample and sources

The sample consists of U.S. domestic equity mutual funds from 1994 to 2000. The core source for restrictions is SEC Form N-SAR, matched to fund characteristics, board characteristics, manager data, fund-family information, Morningstar style classifications, and returns.

4.2 Six investment restrictions

The paper studies whether a fund is restricted from:

- borrowing;
- trading on margin;
- short-selling;
- using individual equity options;
- using equity index futures;
- holding restricted securities.

Interpretation: These restrictions are natural candidates for agency-control devices because they limit leverage, downside risk, derivatives exposure, illiquidity, and deviations from declared investment style.

4.3 Constraint score

The overall constraint score is a summary index of the six restrictions:

$$Score_{it} = \frac{1}{6} \sum_{k=1}^6 Restriction_{kit}.$$

A higher score means that a fund faces a more restrictive investment policy.

Interpretation: The score compresses multiple contractual limits into one measure, making it possible to study overall restrictiveness while also examining individual restrictions separately.

4.4 Alternative monitoring variables

- **BoardSize:** number of directors.
- **PropIndep:** fraction of independent directors.
- **Team:** whether the fund is team-managed.
- **MgrAge:** manager experience or tenure proxy.
- **FrontLoad / BackLoad:** sales-charge structure.
- **Top 10:** indicator for whether the fund belongs to one of the largest fund complexes.

Interpretation: These variables proxy for substitute monitoring mechanisms. If monitoring and explicit constraints substitute for one another, constraints should be more common when alternative monitoring is weaker.

4.5 Performance measures

To test whether restrictions impose performance costs, the paper constructs long-short portfolios based on constraint scores and regression residuals, then examines raw returns and Fama-French abnormal returns.

5 Empirical specifications

5.1 Constraint-score regressions

The main regression relates the overall constraint score to monitoring variables and controls:

$$Score_{it} = \alpha + \beta' Monitoring_{it} + \gamma' Controls_{it} + \delta_{Style} + \varepsilon_{it}.$$

The authors estimate pooled OLS, random effects, between estimators, and double-censored Tobit models.

Interpretation: A positive coefficient means that the variable is associated with more restrictive investment policies. A negative coefficient means that the variable is associated with more managerial flexibility.

5.2 Individual-restriction probits

For each of the six restrictions, the paper estimates pooled probit models:

$$Pr(Restriction_{kit} = 1) = \Phi(\alpha_k + \beta'_k Monitoring_{it} + \gamma'_k Controls_{it}).$$

Interpretation: These models ask whether the aggregate results are driven by one restriction or whether the monitoring variables predict many different types of constraints.

5.3 Performance tests

The paper forms long-short strategies based on:

1. the constraint score itself;
2. the absolute residual from the constraint-score regression.

Returns are evaluated within Morningstar style categories and with Fama-French factor regressions.

Interpretation: If constraints are optimally assigned, high-constraint funds should not systematically underperform low-constraint funds after adjusting for style and risk.

6 Identification and empirical design

6.1 Nature of identification

The paper is not a randomized causal design. It is a contracting-pattern test. The identifying variation comes from differences across funds and time in investment restrictions, board structure, manager structure, fund-family monitoring, load fees, and controls.

6.2 Optimal-contracting prediction

The optimal-contracting view predicts that explicit restrictions should be more likely when other monitoring mechanisms are weaker. For example:

- more independent boards should substitute for constraints;
- large fund complexes should substitute for constraints through peer monitoring and reputation;
- team management and experienced managers can reduce labor-market discipline, making restrictions more valuable;
- load charges might proxy for product-market monitoring, though the evidence here is weak.

6.3 Performance implication

A strong test of the optimal-contracting view is that constrained funds should not perform worse. If constraints are chosen efficiently, any loss of flexibility should be offset by reduced agency costs.

6.4 Main threats

- Constraints may be chosen by fund type rather than agency costs.
- N-SAR measures may not capture informal restrictions or actual portfolio behavior.

- Board structure and fund family size may be endogenous to fund governance quality.
- Performance tests may have limited power because return differences are noisy.

Interpretation: The paper addresses these concerns through controls, style categories, multiple estimation methods, individual restriction probits, and return-based validation tests.

7 Tables and figures

7.1 Table 1: Incidence of constrained and unconstrained funds; Table 2: Use of investment techniques by unconstrained funds

Read:

- Table 1 shows that restrictions are widespread but heterogeneous. Margin restrictions are almost universal, while borrowing, short-sale, futures, options, and restricted-security constraints vary more.
- Constraint scores decline over time, especially for newer funds.
- Table 2 shows that even when funds are allowed to use techniques, actual use is often low.
- The gap between permission and usage suggests that constraints do not merely describe actual behavior; they define allowable managerial discretion.

Economic message: The pair of tables separates legal/contractual freedom from actual portfolio choices. Many funds are constrained, and many unconstrained funds still choose not to use all available techniques.

Detailed interpretation: Table 1 is the descriptive core of the paper. It shows that the menu of restrictions is neither random nor uniform. Table 2 matters because it prevents a simple interpretation that “unconstrained” funds are necessarily aggressive users of short sales, derivatives, or leverage.

Table 1

Incidence of constrained and unconstrained funds

This table provides summary statistics on the incidence of investment restrictions in the sample of domestic equity funds. The table shows the proportion of funds that reported they were not permitted to engage in the practices listed in the column heads. Options refers to the use of individual equity options, Futures to the use of index futures contracts, Restricted to the holding of restricted stock in the fund portfolio, and Borrow, Margin, and Short to whether or not the fund is allowed to borrow, trade on margin, or short sell. Score is the overall investment restriction defined in Section 2.2.2. Panel A describes the results for all funds and Panel B divides the funds into age cohorts.

Year	<i>N</i>	Borrow	Margin	Short	Options	Futures	Restricted	Score
<i>Panel A: All funds, by year</i>								
1994	679	0.212	0.931	0.733	0.290	0.409	0.222	0.399
1995	1,050	0.235	0.916	0.724	0.304	0.409	0.239	0.407
1996	1,201	0.230	0.910	0.700	0.277	0.379	0.209	0.384
1997	1,411	0.219	0.915	0.671	0.247	0.357	0.187	0.364
1998	1,605	0.240	0.918	0.689	0.241	0.352	0.165	0.358
1999	1,741	0.224	0.910	0.684	0.230	0.330	0.144	0.342
2000	1,838	0.206	0.893	0.661	0.227	0.314	0.149	0.335
<i>Overall</i>	9,525	0.224	0.911	0.689	0.252	0.355	0.179	0.363
<i>Panel B: Funds divided by cohort and by year</i>								
<i>Cohort 1: Fund inception prior to 1980</i>								
1994	131	0.252	0.939	0.809	0.366	0.489	0.221	0.439
1995	187	0.310	0.941	0.775	0.388	0.487	0.235	0.450
1996	188	0.303	0.931	0.745	0.381	0.505	0.213	0.439
1997	191	0.288	0.916	0.712	0.318	0.461	0.173	0.401
1998	199	0.367	0.930	0.774	0.345	0.442	0.186	0.424
1999	193	0.337	0.912	0.725	0.258	0.373	0.130	0.368
2000	194	0.320	0.912	0.696	0.263	0.366	0.180	0.379
<i>Overall</i>	1,283	0.314	0.925	0.745	0.329	0.443	0.189	0.413
<i>Cohort 2: Fund inception from 1980 to 1990</i>								
1994	224	0.192	0.960	0.749	0.304	0.478	0.259	0.429
1995	302	0.199	0.950	0.737	0.312	0.492	0.256	0.430
1996	298	0.219	0.943	0.740	0.293	0.428	0.222	0.406
1997	289	0.205	0.944	0.735	0.274	0.431	0.198	0.394
1998	298	0.243	0.946	0.740	0.250	0.395	0.182	0.383
1999	285	0.208	0.923	0.708	0.222	0.363	0.155	0.353
2000	292	0.193	0.907	0.652	0.214	0.362	0.141	0.338
<i>Overall</i>	1,988	0.209	0.938	0.722	0.266	0.420	0.201	0.389
<i>Cohort 3: Fund inception after 1990</i>								
1994	324	0.210	0.907	0.691	0.250	0.330	0.198	0.364
1995	561	0.228	0.889	0.701	0.271	0.339	0.232	0.381
1996	715	0.215	0.891	0.672	0.243	0.326	0.203	0.360
1997	931	0.209	0.905	0.642	0.223	0.314	0.187	0.347
1998	1,108	0.216	0.908	0.660	0.220	0.324	0.156	0.340
1999	1,263	0.211	0.907	0.673	0.227	0.316	0.143	0.336
2000	1,352	0.192	0.888	0.659	0.225	0.296	0.146	0.328
<i>Overall</i>	6,254	0.209	0.899	0.666	0.232	0.317	0.170	0.345

(a) Table 1: Incidence of constrained and unconstrained funds.

Table 2

Use of investment techniques by unconstrained funds

This table provides univariate summary statistics on the actual use of different investment practices by funds that are permitted to engage in these practices. Options refers to the use of individual equity options, Futures to the use of index futures contracts, Restricted to the holding of restricted stock in the fund portfolio, and Borrow, Margin, and Short to whether or not the fund is allowed to borrow, trade on margin, or short sell. Panel A describes the results for all funds and Panel B divides the funds into age cohorts.

Year	Borrow	Margin	Short	Options	Futures	Restricted
<i>Panel A: All funds, by year</i>						
1994	0.099	0.128	0.110	0.124	0.147	0.234
1995	0.076	0.056	0.102	0.118	0.130	0.227
1996	0.072	0.046	0.117	0.085	0.130	0.220
1997	0.081	0.025	0.101	0.090	0.145	0.213
1998	0.099	0.008	0.089	0.104	0.148	0.186
1999	0.113	0.013	0.089	0.089	0.142	0.153
2000	0.121	0.005	0.097	0.086	0.163	0.178
<i>Overall</i>	0.097	0.027	0.098	0.096	0.145	0.194
<i>Panel B: Funds divided by cohort and by year</i>						
<i>Cohort 1: Fund inception prior to 1980</i>						
1994	0.122	0.000	0.000	0.193	0.194	0.412
1995	0.070	0.000	0.071	0.191	0.188	0.399
1996	0.076	0.000	0.083	0.162	0.161	0.412
1997	0.081	0.000	0.109	0.176	0.223	0.418
1998	0.095	0.000	0.133	0.168	0.198	0.383
1999	0.109	0.000	0.094	0.104	0.182	0.310
2000	0.167	0.000	0.085	0.140	0.211	0.321
<i>Overall</i>	0.102	0.000	0.089	0.159	0.195	0.376
<i>Cohort 2: Fund inception prior to 1980 to 1990</i>						
1994	0.127	0.111	0.107	0.115	0.120	0.283
1995	0.091	0.000	0.127	0.106	0.144	0.272
1996	0.073	0.000	0.117	0.110	0.141	0.316
1997	0.083	0.000	0.079	0.096	0.152	0.312
1998	0.089	0.000	0.065	0.131	0.134	0.248
1999	0.129	0.000	0.084	0.154	0.155	0.204
2000	0.120	0.000	0.050	0.149	0.184	0.225
<i>Overall</i>	0.101	0.008	0.087	0.124	0.149	0.264
<i>Cohort 3: Fund inception after 1990</i>						
1994	0.070	0.167	0.140	0.107	0.143	0.135
1995	0.069	0.081	0.095	0.105	0.111	0.153
1996	0.071	0.064	0.124	0.059	0.120	0.132
1997	0.080	0.034	0.105	0.073	0.130	0.140
1998	0.101	0.010	0.088	0.088	0.144	0.136
1999	0.110	0.017	0.090	0.073	0.133	0.117
2000	0.116	0.007	0.108	0.065	0.153	0.148
<i>Overall</i>	0.095	0.035	0.103	0.077	0.136	0.136

(b) Table 2: Use of investment techniques by unconstrained funds.

The first pair of tables documents both contractual restrictions and actual use of investment techniques.

7.2 Table 3: Pooled regression estimates

Read:

- Table 3 links the constraint score to governance and monitoring variables.
- PropIndep is negative and significant: more independent boards are associated with fewer restrictions.
- Team and MgrAge are positive and significant: team-managed funds and more experienced managers face more restrictions.
- Top 10 is negative and significant: funds in large complexes face fewer restrictions.
- Load variables are weak and mostly insignificant.

Economic message: The regression evidence supports a substitution view of governance. Independent boards and large fund complexes substitute for explicit constraints, while weaker or less direct labor-market monitoring is associated with more constraints.

Detailed interpretation: The most important coefficients are PropIndep and Top 10. Both are negative, consistent with explicit constraints being less necessary when the fund has stronger governance or more peer/reputation monitoring. The positive Team and MgrAge coefficients suggest that constraints become more useful when individual accountability or career discipline is weaker.

Table 3

Pooled regression estimates

This table reports estimated parameters for four different pooled observation models. The dependent variable is score, the overall measure of fund constraint constructed in Section 2.2.2. *BoardSize*, *PropIndep*, *Team*, *MgrAge*, *Backload*, *Frontload*, and *Top 10* are the explanatory variables related to the different contracting hypotheses for the use of direct investment restrictions. *LogFundAge*, *Turnover*, and *LogTNA* are control variables, and LB through SG are indicator variables for Morningstar style categories that are also included as controls. The different columns correspond to different estimation methods based on pooling all of the fund-year data. The first column is the pooled OLS estimator. The second column is the random-effects (R.E.) model. The third column is the “between” OLS cross-sectional estimator constructed from the time-series averages. The final column is a double-censored Tobit estimator that explicitly accounts for the truncated nature of score and uses the information in the probability mass at 0 and 1. P-values are reported in parentheses underneath the regression coefficients. Summary statistics for the regressions are reported in the last three rows. The F - or χ^2 -statistic tests for the joint significance of all the monitoring variables.

Variable	Pred. sign	Pooled OLS	R.E. (GLS)	Between-estimator	Panel Tobit
Monitoring hypothesis					
<i>BoardSize</i>	+	−0.003 (0.093)	−0.001 (0.419)	−0.003 (0.214)	− 0.002 (0.047)
<i>PropIndep</i>	−	− 0.217 (0.000)	− 0.073 (0.000)	− 0.239 (0.000)	− 0.105 (0.000)
<i>Team</i>	+	0.030 (0.001)	0.010 (0.047)	0.033 (0.004)	0.020 (0.001)
<i>MgrAge</i>	+	0.005 (0.000)	0.002 (0.007)	0.005 (0.000)	0.003 (0.000)
<i>FrontLoad</i>	+	−0.004 (0.178)	−0.015 (0.060)	−0.006 (0.657)	−0.005 (0.484)
<i>BackLoad</i>	+	−0.019 (0.697)	−0.016 (0.079)	−0.017 (0.306)	−0.011 (0.315)
<i>Top 10</i>	−	− 0.039 (0.009)	− 0.059 (0.000)	− 0.049 (0.016)	− 0.063 (0.000)
Control variables					
<i>LogFundAge</i>		0.060 (0.000)	0.002 (0.719)	0.066 (0.000)	0.023 (0.000)
<i>LogTNA</i>		− 0.028 (0.000)	− 0.012 (0.000)	− 0.027 (0.000)	− 0.014 (0.000)
<i>Turnover</i>		− 0.016 (0.029)	− 0.005 (0.001)	− 0.019 (0.000)	− 0.005 (0.002)
<i>LB</i>		0.001 (0.932)	0.000 (0.963)	−0.008 (0.707)	−0.007 (0.279)
<i>LG</i>		−0.011 (0.514)	−0.005 (0.577)	−0.013 (0.540)	−0.004 (0.640)
<i>MV</i>		−0.016 (0.309)	0.004 (0.625)	−0.035 (0.170)	0.008 (0.395)
<i>MB</i>		−0.018 (0.298)	−0.008 (0.409)	−0.035 (0.299)	−0.004 (0.667)
<i>MG</i>		− 0.037 (0.019)	− 0.024 (0.012)	−0.036 (0.124)	− 0.022 (0.032)
<i>SV</i>		−0.015 (0.442)	0.010 (0.394)	0.009 (0.698)	0.023 (0.045)
<i>SB</i>		0.008 (0.665)	−0.002 (0.875)	0.018 (0.200)	0.009 (0.470)

(a) Table 3: Pooled regression estimates, main coefficients.

Table 3 (continued)

Variable	Pred. sign	Pooled OLS	R.E. (GLS)	Between-estimator	Panel Tobit
<i>SG</i>		−0.009 (0.610)	−0.011 (0.292)	−0.020 (0.427)	−0.007 (0.543)
<i>Constant</i>		0.540 (0.000)	0.468 (0.000)	0.541 (0.000)	0.475 (0.000)
# of Funds		7,259	7,259	7,259	7,259
F or χ^2		9.19 (F) (0.000)	65.37 (χ^2) (0.000)	9.97 (F) (0.000)	91.09 (χ^2) (0.000)
R^2		0.107	0.053	0.107	n.a.

(b) Table 3 continued: sample size, joint tests, and fit.

Table 3 estimates how monitoring mechanisms predict the overall restriction score.

7.3 Table 4: Summary tests from individual constraint pooled probits

Read:

- Panel A shows that the covariates jointly predict each individual restriction.
- Panel B shows that PropIndep, Team, MgrAge, and Top 10 are jointly significant across restrictions.
- FrontLoad and BackLoad are not robustly significant.

Economic message: The main regression results are not an artifact of averaging across unrelated restrictions. The monitoring variables predict individual constraints as well.

Detailed interpretation: Table 4 strengthens the interpretation of Table 3. It shows that governance and monitoring variables matter across several types of restrictions, not just because one specific constraint happens to dominate the composite score.

This table reports the summary statistics for tests based on coefficients from pooled probit models of the individual fund restrictions. There are 7,259 funds in the samples used for each regression. The dependent variable equals one if the particular investment practice is prohibited and zero otherwise. *BoardSize*, *PropIndep*, *Team*, *MgrAge*, *Backload*, *Frontload*, and *Top10* are the explanatory variables related to the different contracting hypotheses for the use of direct investment restrictions. *LogFundAge*, *Turnover*, and *LogTNA* are control variables, and LB through SG are indicator variables for Morningstar style categories that are also included as controls. *P*-values – based on robust standard errors – are reported in parentheses underneath the regression coefficients. The χ^2 -statistic tests for the joint significance of all the monitoring variables in a given equation is reported in Panel A. Panel B reports joint tests of the hypotheses that all of the coefficients equal zero (asymptotically distributed χ^2 with six degrees of freedom) and that the average coefficient value equals zero (asymptotically distributed χ^2 with one degree of freedom).

	Options	Futures	Restricted	Borrowing	Margin	Short	
<i>Panel A: Joint significance of coefficients by constraint type</i>							
χ^2 test statistic	44.32 (0.000)	52.77 (0.000)	70.39 (0.000)	21.65 (0.003)	17.43 (0.015)	30.26 (0.000)	
	<i>BoardSize</i>	<i>PropIndep</i>	<i>Team</i>	<i>MgrAge</i>	<i>FrontLoad</i>	<i>BackLoad</i>	<i>Top 10</i>
<i>Panel B: Joint significance of monitoring and control variables across constraints</i>							
All zero	31.28 (0.000)	54.11 (0.000)	17.53 (0.000)	33.11 (0.000)	10.39 (0.109)	10.90 (0.092)	32.25 (0.000)
Avg. zero	2.40 (0.121)	15.96 (0.000)	12.46 (0.000)	9.70 (0.002)	0.62 (0.429)	1.28 (0.257)	5.26 (0.022)

Table 4: Summary tests from individual constraint pooled probits. The single table is shown at width 0.6 linewidth under the one-visual rule.

7.4 Table 5: Returns to a long-short strategy based on score; Table 6: Returns to a long-short strategy based on absolute errors

Read:

- Table 5 tests whether high-constraint funds underperform low-constraint funds.
- Table 6 tests whether funds whose constraint levels deviate from model predictions underperform.
- Neither strategy produces economically or statistically significant abnormal returns.
- The Fama-French alphas are generally close to zero and insignificant.

Economic message: The performance results support optimal contracting. Constraints do not appear to destroy risk-adjusted value, and deviations from predicted constraint levels do not generate exploitable return patterns.

Detailed interpretation: These tables are the performance validation tests. If constraints were simply costly limits on manager skill, low-constraint funds should outperform. The absence of significant long-short returns suggests that the market and fund boards choose constraints in a way that balances benefits and costs.

Table 5

Summary statistics on the returns to a long-short strategy based on score

This table presents returns to a zero net investment strategy formed by going long in funds whose *Score* value is lower than the average – either overall or within a Morningstar style category – and short funds whose *Score* value is above the average. Weights in the long and short portfolios are proportional to the distance of a fund's *Score* value from the average. Investment style refers to Morningstar categories, where L is 'large-cap', M is 'mid-cap', S is 'small-cap'. V is 'value', B is 'blend', G is 'growth', and LV (for example) is 'a large-cap value' portfolio. The Fama-French factors are: *MF* (excess market return), *SMB* (the returns to a zero investment portfolio that is long a small-cap portfolio and short a large-cap portfolio), and *HML* (the returns to a zero investment portfolio that is long a portfolio of high B/M stocks and short a portfolio of low B/M stocks). *P*-values based on robust standard errors (Newey-West estimator with six lags) are reported in parentheses beneath the Fama-French abnormal returns (α) and the Fama-French factor loadings ($\beta_{MF}, \beta_{SMB}, \beta_{HML}$).

Investment style	Returns		Fama-French regressions				R^2
	Mean	Std. Dev.	α	β_{MF}	β_{SMB}	β_{HML}	
<i>LV</i>	-0.00115	0.00448	-0.00046 (0.125)	-0.078 (0.000)	0.025 (0.002)	-0.019 (0.024)	0.61
<i>LB</i>	-0.00060	0.00619	0.00026 (0.579)	-0.094 (0.000)	-0.008 (0.376)	0.011 (0.435)	0.61
<i>LG</i>	0.00079	0.00786	0.00134 (0.140)	-0.059 (0.046)	-0.020 (0.171)	0.021 (0.253)	0.24
<i>MV</i>	-0.00064	0.00572	0.00013 (0.79)	-0.085 (0.000)	-0.004 (0.724)	-0.043 (0.003)	0.41
<i>MB</i>	0.00012	0.00985	0.00059 (0.632)	-0.054 (0.136)	0.009 (0.699)	-0.106 (0.000)	0.27
<i>MG</i>	-0.00015	0.00685	0.00041 (0.597)	-0.062 (0.035)	-0.004 (0.843)	-0.019 (0.311)	0.15
<i>SV</i>	-0.00146	0.00758	-0.00091 (0.319)	-0.059 (0.046)	-0.056 (0.010)	-0.075 (0.001)	0.23
<i>SB</i>	0.00190	0.01242	0.00252 (0.166)	-0.072 (0.270)	0.032 (0.372)	-0.091 (0.005)	0.17
<i>SG</i>	0.00065	0.01091	0.00017 (0.899)	0.054 (0.141)	-0.009 (0.840)	-0.021 (0.501)	0.08
Overall	-0.00041	0.00410	0.00003 (0.911)	-0.050 (0.000)	0.004 (0.717)	-0.063 (0.000)	0.59

(a) Table 5: Return strategy based on constraint score.

Table 6

Summary statistics on the returns to a long-short strategy based on absolute errors in the regression of score on the economic determinants of constraint use

This table presents returns to a zero net investment strategy formed by going long in funds with small absolute regression errors in the *Score* regressions financed by short positions in funds with large absolute regression errors. We repeat these strategies both overall and within a Morningstar style category. Weights in the long and short portfolios are proportional to the size of the absolute errors. Investment style refers to Morningstar categories, where L is 'large-cap', M is 'mid-cap', S is 'small-cap', V is 'value', B is 'blend', G is 'growth', and LV (for example) is 'a large-cap value' portfolio. The Fama-French factors are: *MF* (excess market return), *SMB* (the returns to a zero investment portfolio that is long a small-cap portfolio and short a large-cap portfolio), and *HML* (the returns to a zero investment portfolio that is long a portfolio of high B/M stocks and short a portfolio of low B/M stocks). *P*-values based on robust standard errors (Newey-West estimator with six lags) are reported in parentheses beneath the Fama-French abnormal returns (α) and the Fama-French factor loadings ($\beta_{MF}, \beta_{SMB}, \beta_{HML}$).

Investment style	Returns		Fama-French regressions				R^2
	Mean	Std. Dev.	α	β_{MF}	β_{SMB}	β_{HML}	
<i>LV</i>	-0.00143	0.00488	-0.00063 (0.058)	-0.091 (0.000)	0.018 (0.005)	-0.010 (0.030)	0.69
<i>LB</i>	0.00021	0.00553	0.00071 (0.174)	-0.055 (0.005)	-0.025 (0.030)	0.005 (0.637)	0.33
<i>LG</i>	-0.00106	0.00945	-0.00039 (0.693)	-0.071 (0.000)	-0.016 (0.372)	0.067 (0.000)	0.42
<i>MV</i>	0.00026	0.00862	0.00076 (0.588)	-0.055 (0.042)	-0.002 (0.929)	0.004 (0.758)	0.10
<i>MB</i>	0.00058	0.00819	0.00071 (0.464)	-0.013 (0.890)	-0.019 (0.577)	0.005 (0.613)	0.02
<i>MG</i>	-0.00027	0.00886	0.00027 (0.788)	-0.059 (0.078)	-0.013 (0.597)	0.011 (0.547)	0.14
<i>SV</i>	0.00039	0.00802	0.00119 (0.181)	-0.087 (0.000)	-0.040 (0.032)	-0.053 (0.011)	0.25
<i>SB</i>	-0.00044	0.01008	0.00019 (0.820)	-0.066 (0.003)	-0.069 (0.025)	-0.055 (0.021)	0.14
<i>SG</i>	0.00223	0.01104	0.00190 (0.255)	0.042 (0.303)	-0.061 (0.061)	0.035 (0.310)	0.11
Overall	-0.00032	0.00421	0.00015 (0.808)	-0.052 (0.000)	-0.005 (0.686)	-0.023 (0.002)	0.28

(b) Table 6: Return strategy based on absolute regression errors.

Tables 5 and 6 test whether constraint intensity or constraint-model deviations predict fund performance.

7.5 Table 7: Regression of fund returns on constraint variables

Read:

- Table 7 regresses fund returns on predicted restriction levels and residual restriction levels.
- The coefficients do not provide evidence that constraints lower performance.
- The monitoring and control variables do not overturn the conclusion from Tables 5 and 6.

Economic message: The final return test confirms that constraint use is not associated with economically meaningful underperformance.

Detailed interpretation: Table 7 is a regression-based counterpart to the portfolio strategy tests. It shows that once constraints are decomposed into predicted and residual components, neither component strongly explains fund returns.

Table 7

Fund level cross-sectional regressions of Fama-French α on score, monitoring, and control variables. This table reports results from the regression of the alpha from the Fama-French three-factor model, estimated using one year of monthly observations, on *Score* (or *Score*, monitoring, and control variables) measured at the beginning of the period. The supplementary variables are described in detail in Section 4. They include governance measures, career concerns, load fee, peer monitoring, liquidity measures, fund size and age, turnover, and Morningstar style categories. In the interest of saving space, the coefficient estimates for the control variables are not reported. The omitted Morningstar category is large-cap value. The first column is the pooled OLS estimator. The second column is the random-effects (R.E.) model. The third column is the “between” OLS cross-sectional estimator constructed from the time-series averages. The fourth column reports the average coefficient from yearly regressions. *P*-values are reported in parentheses, based on robust standard errors for all the models and on the Fama-MacBeth procedure for the averages.

	Pooled OLS	R.E. (GLS)	Between estimator	Fama-MacBeth
<i>Panel A: Regressions with score alone</i>				
Score	−0.010 (0.819)	−0.021 (0.676)	0.023 (0.719)	0.000 (0.997)
Constant	−0.089 (0.000)	−0.079 (0.001)	−0.093 (0.001)	−0.117 (0.251)
<i>Panel B: Regressions with score, monitoring, and control variables</i>				
Score	0.011 (0.832)	−0.051 (0.402)	0.108 (0.160)	0.011 (0.873)
BoardSize	−0.001 (0.899)	0.005 (0.373)	0.008 (0.268)	−0.006 (0.227)
PropIndep	−0.092 (0.359)	0.023 (0.836)	−0.247 (0.111)	−0.161 (0.072)
Team	0.034 (0.173)	0.052 (0.065)	0.012 (0.760)	0.011 (0.575)
MgrAge	0.008 (0.000)	0.009 (0.007)	0.009 (0.029)	0.008 (0.019)
FrontLoad	−0.192 (0.000)	−0.186 (0.000)	−0.241 (0.000)	−0.124 (0.148)
BackLoad	−0.166 (0.000)	−0.146 (0.002)	−0.214 (0.000)	−0.148 (0.100)
Top 10	0.076 (0.026)	0.046 (0.393)	0.029 (0.676)	0.110 (0.061)
Constant	0.045 (0.639)	0.105 (0.927)	−0.049 (0.725)	0.052 (0.251)

report them in the last column.¹² None of the four models provides evidence of a significant relation between the score measure and Fama-French alphas.

Panel B of Table 7 reports the estimated coefficients with the full set of monitoring and control variables also included in the regression. These regressions confirm that, holding other fund characteristics constant, there is no effect of the prior level of

¹²For each coefficient in the last column, we compute a test statistic of the form $\hat{\delta}/s.e.(\hat{\delta})$ which is *t*-

Table 7: Regression of fund returns on constraint variables. The single table is displayed at width 0.6 linewidth under the one-visual rule.

8 Short summary

- Investment restrictions are common in domestic equity mutual funds.
- Restrictions vary across constraint types, fund age cohorts, and calendar time.
- Many unconstrained funds do not use all permitted investment techniques.
- Constraint scores are lower when boards are more independent and when funds belong to large complexes.
- Constraint scores are higher for team-managed funds and more experienced managers.
- Load-fee structure does not strongly explain restrictions.
- Individual restriction probits confirm that monitoring variables matter across specific constraints.
- High-constraint funds do not underperform low-constraint funds on a risk-adjusted basis.
- The evidence supports an optimal-contracting view rather than a simple managerial-restriction cost view.

9 Conclusion

9.1 Core conclusion

Almazan, Brown, Carlson, and Chapman show that mutual fund investment restrictions are systematic contractual choices. Constraints are not randomly assigned and are not merely bureaucratic limits. They are related to the governance and monitoring environment of the fund.

9.2 Economic intuition

Delegated portfolio management creates agency problems because managers can take actions that investors may not want or cannot perfectly monitor. Explicit restrictions reduce the feasible action space. They are most useful when other forms of monitoring are weaker and less useful when independent boards or large fund complexes provide alternative oversight.

9.3 Incremental contribution revisited

The paper contributes by bringing contract design into the mutual fund literature. It shows that the structure of investment policy statements is an economically meaningful governance mechanism, similar to covenants in debt contracts.

9.4 Performance implications

The paper finds no evidence that constrained funds earn lower abnormal returns. This is important because it means constraints need not be interpreted as costly handcuffs. They may instead be chosen to minimize total agency and monitoring costs.

9.5 Limitations

The paper observes formal restrictions, not all informal monitoring or actual manager discretion. The performance tests may have limited power, and the assignment of constraints is endogenous. The paper's interpretation is therefore best viewed as evidence consistent with optimal contracting, rather than definitive causal proof.

9.6 Takeaway

The central lesson is that constraints can be productive contractual devices. In mutual funds, more freedom is not always better. The optimal degree of managerial discretion depends on the surrounding governance, monitoring, and organizational environment.